Wine Quality Prediction Using XGBoost and LightGBM with Hyperparameter Tuning

Machine Learning (25CS2103E)

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ABSTRACT

Wine quality prediction is an important machine learning application that helps estimate the quality of wine based on its physicochemical properties. This project, “Wine Quality Prediction Using XGBoost and LightGBM with Hyperparameter Tuning,” aims to develop and compare two advanced gradient-boosting machine learning algorithms, XGBoost and LightGBM, for predicting wine quality. The dataset contains various chemical characteristics of wine, such as acidity, residual sugar, chlorides, sulphates, density, pH, and alcohol content.

The project involves data preprocessing, exploratory data analysis, feature selection, and splitting the dataset into training and testing sets. XGBoost and LightGBM models are trained on the processed data, followed by hyperparameter tuning to optimize their performance. The models are evaluated using appropriate performance metrics, and their results are compared to identify the better-performing model. The proposed system demonstrates how machine learning can be used to analyze wine characteristics and predict quality efficiently, while also highlighting the advantages of optimized gradient-boosting techniques.

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